

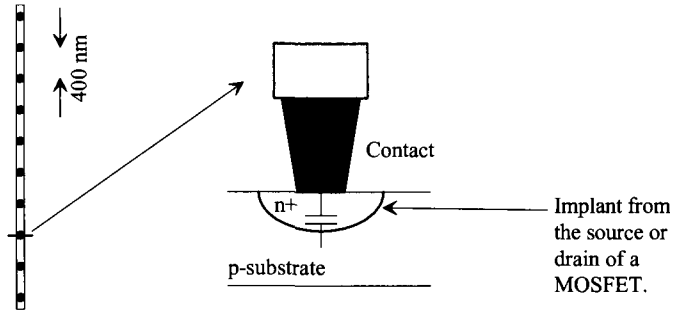


Figure 16.3 How a bit line is loaded with implants (depletion capacitance).

16.1.1 Sensing Basics

In this section we discuss sensing the datum from a memory cell. Examine Fig. 16.4. When a memory cell is accessed (the word [row] line goes high), the datum from the cell (a charge) is placed on the bit line. The bit line voltage changes. At this point we know the bit line looks like a capacitor and that only one word line can go high at a time in a memory array (so that we don't have two memory cells trying to dump their data onto the same bit line at the same time). The voltage movement on the bit line, ΔV_{bit} , may be very small, e.g., 50 mV or less, and so determining if the voltage is moving upwards or downwards can be challenging. In addition, we would like our sense amplifier to drive the bit line to full, valid, logic levels (for speed reasons in some memories and to refresh the cell in a dynamic RAM, DRAM). Let's consider some sense amplifier topologies.

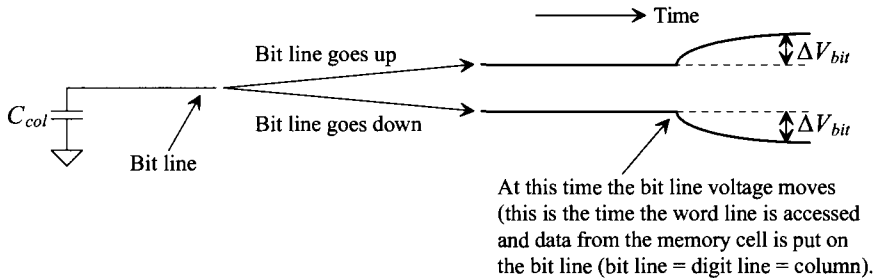


Figure 16.4 Sensing a change on the bit line in an array.

NMOS Sense Amplifier (NSA)

Consider the schematic of an NMOS sense amplifier (NSA) seen in Fig. 16.5. M1 and M2 form the NMOS portion of an inverter-based latch. The idea is to develop an imbalance on the gates of M1/M2 so that one MOSFET turns on harder than the other. Before we start sensing, we set the drains of M1/M2 to the same potential (an equilibrium potential). When *sense_N* goes high (indicating that we are starting the sense operation), the signal *NLAT* (N-latch) goes to ground (the sources of M1/M2 move to ground). While this circuit clearly can't pull the bit line high (we'll add a PMOS sense amplifier later to do this), we should see a problem. How do we get good sensing if the transistor loads are not balanced?

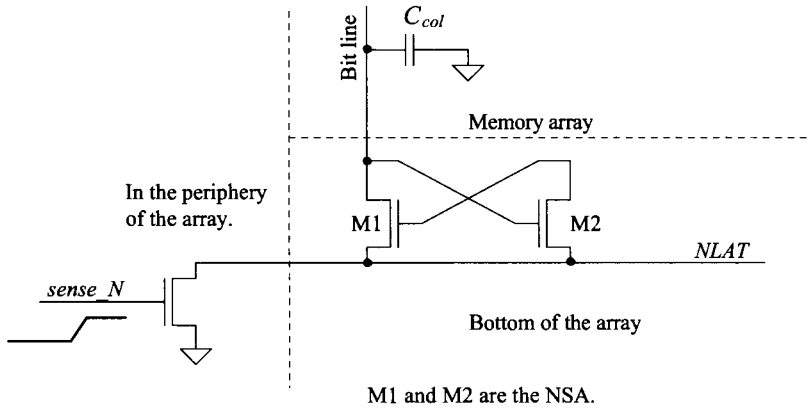


Figure 16.5 Development of an NMOS sense amplifier (NSA).

The Open Array Architecture

To provide the same load capacitance for each input of the NSA, two arrays can be used, as seen in Fig. 16.6. The array architecture in Fig. 16.6, from the side, looks like an open book and so it is called an “open array architecture.” Let’s use some numbers and the partial schematic seen in Fig. 16.7 to illustrate the sense amplifier’s operation.

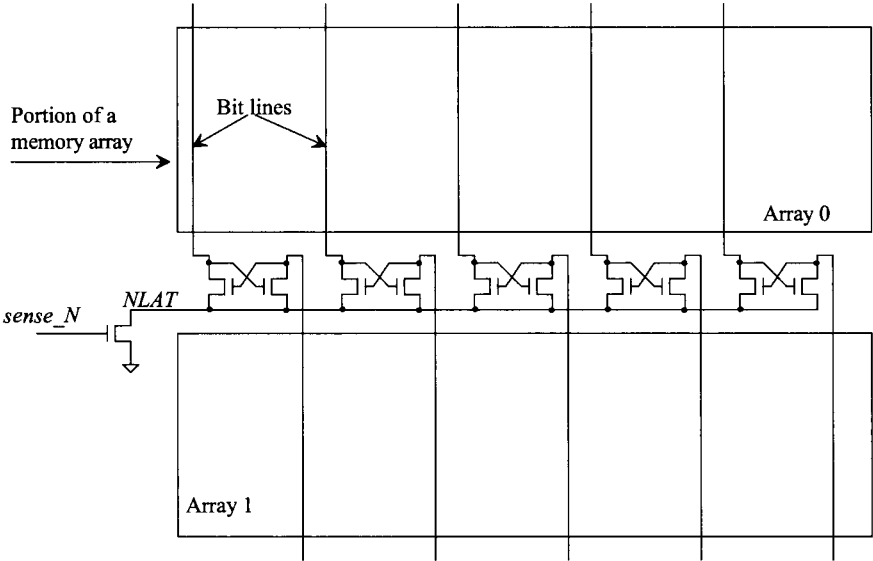


Figure 16.6 How the NSA is placed between two memory arrays in the so-called open memory array architecture.

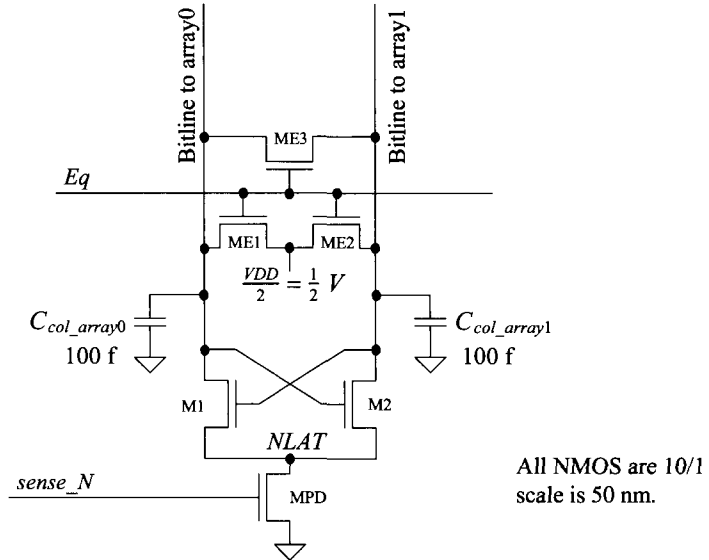


Figure 16.7 An NSA with equilibration circuitry and connections to two bit lines.

In the following we assume $V_{DD} = 1$ V and the column capacitance is 100 fF. We start any sense operation by equilibrating the bit lines (the inputs to the sense amplifier). In Fig. 16.7 ME1–ME3 short the bit lines together and to $V_{DD}/2$ ($= 0.5$ V). When the equilibrate signal, Eq , goes high, all of the word lines are low. We are not accessing any data in the array but, rather, getting ready for the sense (read) operation. Figure 16.8 shows how the Eq signal is asserted for a short period of time and how, during this time, the bit lines are equilibrated together and to 0.5 V.

During a sense operation, one of the bit lines will be pulled from $V_{DD}/2$ to V_{DD} . The other line will be pulled from $V_{DD}/2$ to ground. The amount of power used during a sense depends on frequency and given by

$$P_{avg} = (\text{number of sense amplifiers}) \cdot C_{col} \cdot (V_{DD}/2)^2 \cdot f \quad (16.4)$$

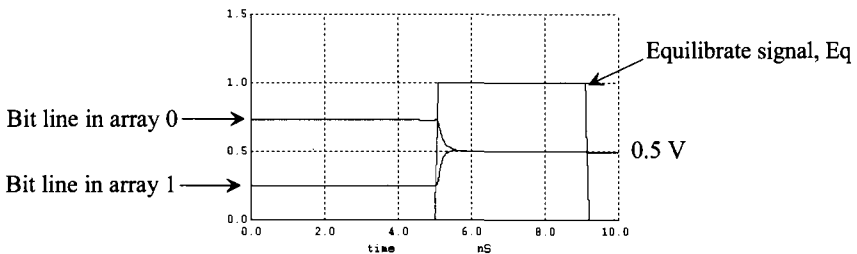


Figure 16.8 How the equilibrate circuitry operates.

Note, however, that when we equilibrate, no power is consumed. The two bit line capacitances simply share their charge (one going from ground to $VDD/2$ while the other goes from VDD to $VDD/2$).

To show a sensing operation, let's use the basic one-transistor, one-capacitor (1T1C) DRAM memory cell seen in Fig. 16.9. To fully turn on the access MOSFET, we need to drive the word line to $VDD + V_{THN}$ (with body effect). A typical value for the memory bit capacitance, C_{mbit} , is 20 fF. If the voltage on this capacitance is called V_{mbit} and the bit line is precharged to $VDD/2$, then we can write the total charge on both capacitors *before* the access MOSFET turns on as

$$Q_{tot} = C_{mbit} V_{mbit} + (VDD/2) \cdot C_{col_array} \quad (16.5)$$

After the access MOSFET turns on, the voltage across each capacitor will be the same. We'll call this voltage, V_{final} . Since charge must be conserved

$$V_{final}(C_{mbit} + C_{col_array}) = C_{mbit} V_{mbit} + (VDD/2) \cdot C_{col_array} \quad (16.6)$$

or

$$V_{final} = \frac{C_{mbit} V_{mbit} + (VDD/2) \cdot C_{col_array}}{C_{mbit} + C_{col_array}} \quad (16.7)$$

If a logic one is stored on C_{mbit} (which means V_{mbit} is VDD , or here, 1 V) and $C_{mbit} = 20$ fF and $C_{col_array} = 100$ fF, then $V_{final} = 0.583$ V. The change in the bit line voltage is

$$\Delta V_{bit} = V_{final} - VDD/2 \quad (16.8)$$

or here $\Delta V_{bit} = 83$ mV .

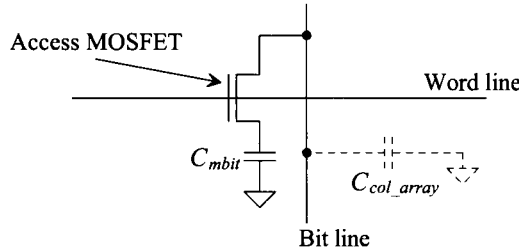


Figure 16.9 The one-transistor, one-capacitor (1T1C) DRAM memory cell.

Figure 16.11 shows the signals resulting from the operation of the sense amplifier in Fig. 16.10. The 1T1C DRAM (mbit) cell in array1 (the DRAM cell at the bottom of the schematic) has its word line held at ground when we sense the data in the cell in array0. The bit line from array1 is simply used as a reference. When we sense the data in array1, the bit line in array0 is used as the reference (and so the word line in array1 is held at ground).

For the simulation results in Fig. 16.11, we start out by equilibrating the bit lines (as in Fig. 16.8). Next our word line, in array0, goes to a voltage greater than $VDD + V_{THN}$. This causes charge sharing between C_{mbit} and the digit line capacitance. For the simulation results seen in Fig. 16.11, we set the voltage on C_{mbit} to zero so we are reading out zero.

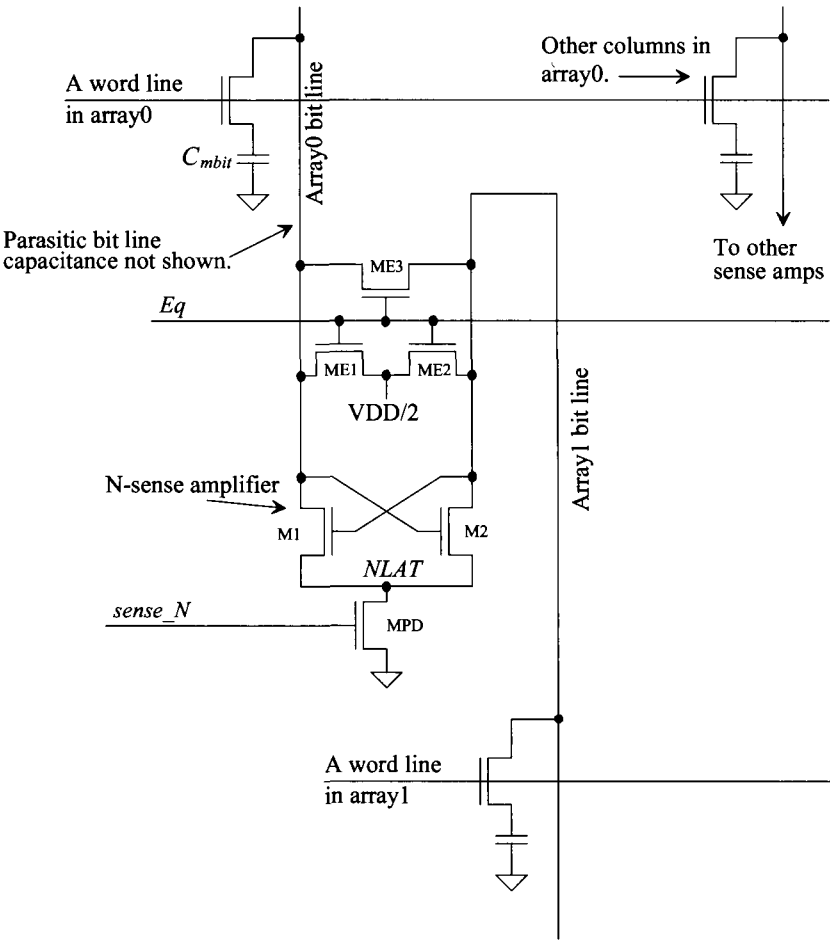


Figure 16.10 The connection of the NSA to the memory arrays.

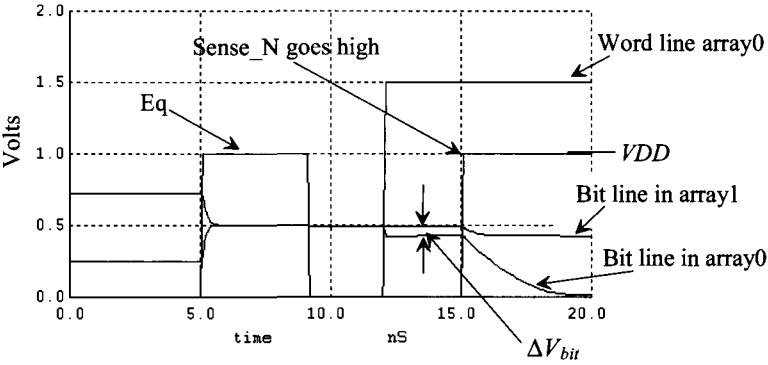


Figure 16.11 N-sense amp's operation.

When *sense_N* goes high, the NSA “fires” causing the bit line in array0 to move to ground. Note that our reference bit line (from array1) droops a little. Figure 16.12 shows the signals in a sensing operation if the mbit cell contains a “1.” The bit line voltage in array0 now increases. The reference bit line in array1 is pulled to ground (this doesn’t harm the data in array1). To pull the bit line in array0 high, we’ll now add a PMOS sense amplifier (PSA).

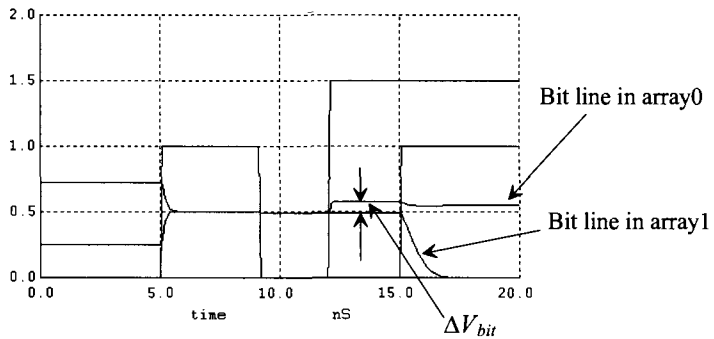


Figure 16.12 Reading out a “1” from the cell in array0.

PMOS Sense Amplifier (PSA)

To pull the bit lines up to *VDD*, we can add a PSA to the periphery of the array. Figure 16.13 shows the schematic of the PSA. The signal *ACT* (active pullup) is common to all of the PSAs on the periphery of the array (as is *NLAT* for the NSAs seen in Fig. 16.6). The signal *sense_P* is active low and indicates that the PSA is firing. The PSA is usually fired after the NSA because the matching of the NMOS devices is, generally, better than

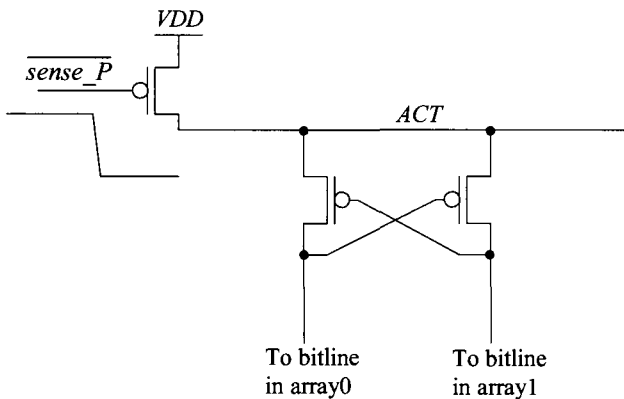


Figure 16.13 Schematic diagram of a PMOS sense amplifier.

the matching of the PMOS. It's not common to fire the sense amplifiers at the same time because of the potentially significant contention current that can flow. Figure 16.14 shows the full operation of a sense when the cell we are reading out has a one stored in it.

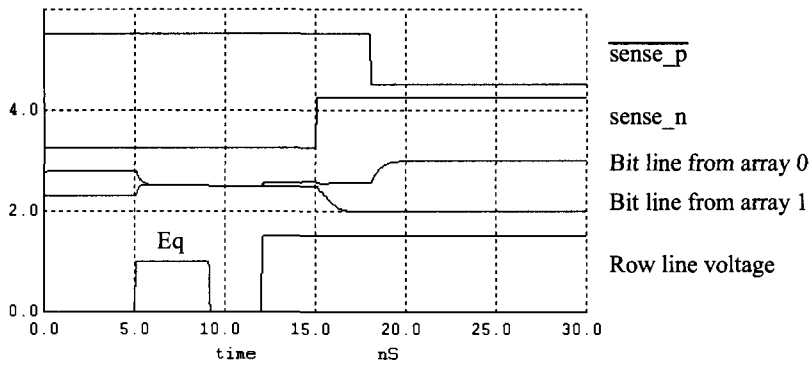


Figure 16.14 How the PSA pulls the bit line from array 0 high.

Refresh Operation

Our sensing operation determined whether a 1 or a 0 was stored in the memory bit. For this particular sense amplifier topology, the inputs and outputs are the same terminals. This is fundamentally important for DRAM operation where the charge can leak off the capacitor because of the finite subthreshold slope of the access transistors or leakage through the MOSFET's source/drain implant to substrate (hence the name “dynamic”). To ensure long-lasting data retention in a DRAM, the cells must be periodically refreshed. When we fire the sense amplifiers, with the access device still conducting, the mbit capacitor is refreshed through the access MOSFET (the bit line is driven to ground or V_{DD} by the sense amp).

16.1.2 The Folded Array

The open array architecture seen in Fig. 16.6 has a memory cell at the location of every intersection of a word line and a bit line. The open array architecture results in the most dense array topology. Unfortunately, because of the physical distance between the bit lines used by the sense amplifier, this architecture is sometimes not used. Figure 16.15 shows the basic problem. Coupled noise (e.g., from the substrate) feeds unequal amounts

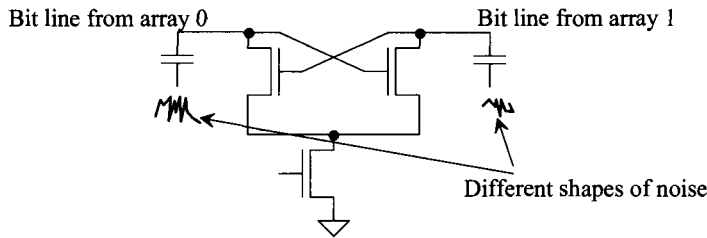


Figure 16.15 Different amplitudes of coupled noise into physically separated bit line.

of charge into the bit lines. This can cause the sense amplifier to make wrong decisions. To attempt to make each bit line see the same sources of noise, we can lay the bit lines out next to each other by folding array 1 on top of array 0 (see Fig. 16.6). The result, called a *folded array architecture*, is seen in Fig. 16.16. The bold lines in this figure are the bit lines from array 1.

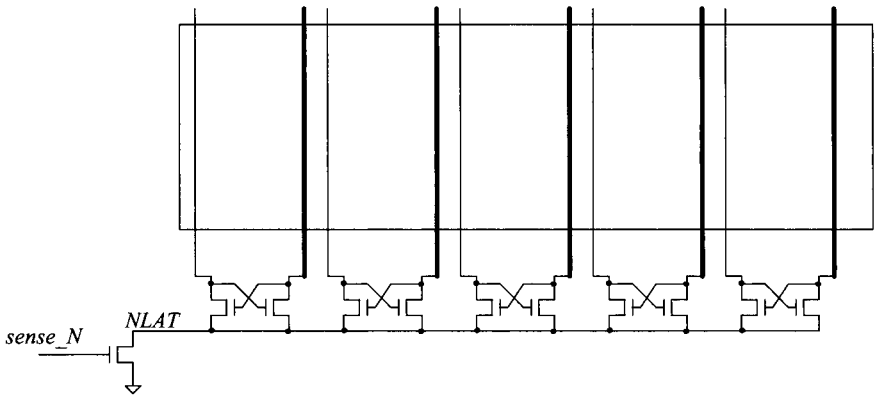


Figure 16.16 The folded array is formed by taking the open array architecture (open book) topology seen in Fig. 16.6 and "closing the book," that is, folding array 1 on top of array 0. Note that the bold lines indicate the bitlines from array 1 in the newly formed array.

What is the cost for this improved noise performance? We know that when a sense amplifier is used, one input is varied by the memory cell we are sensing, while the other input is simply used as a reference. What this means is that instead of having a memory cell at the intersection of every row and column, as in the open array, now we have a memory cell at the location of every other row and column, Fig. 16.17.

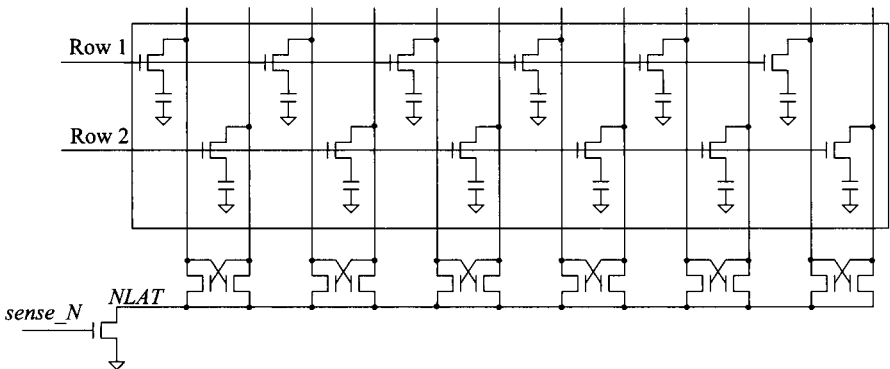


Figure 16.17 How a memory cell is located at every other intersection of a row line and a column line in a folded-area architecture.

Layout of the DRAM Memory Bit (Mbit)

Layout area, especially in a periodic array like a memory, is very important. Reducing the chip size increases the number of die on a wafer. Since the processing cost for a wafer is fixed, having more chips to sell per wafer increases the manufacturer's profit. To reduce the size of an mbit, often (always in DRAM) the contact to the bit line is shared between two mbit cells, Fig. 16.18. The word lines (row lines) are made using silicided polysilicon. Using polysilicon for the word lines can lead to significant delays. As seen in Fig. 16.17, for example, a signal applied to row 1 must propagate down a distributed R (the resistance of the polysilicon) C (the gate oxide capacitance of the MOSFET) line. The propagation delay through a word line can be estimated using

$$t_d = \underbrace{(\text{number of columns}) \cdot \left(WL \frac{\epsilon_{ox}}{t_{ox}} + C_{parasitic} \right)}_{\text{total capacitance on the word line}} \cdot \underbrace{(\text{number of columns}) \cdot R_{gate}}_{\text{total resistance of the word line}} \quad (16.9)$$

The number of columns is simply the number of MOSFETs in the row. The term $WL \frac{\epsilon_{ox}}{t_{ox}}$ is C_{ox} , while R_{gate} is the resistance from one end of the polysilicon gate to the other end in the memory cell layout seen in Fig. 16.18. The term $C_{parasitic}$ is the parasitic capacitance associated with the cell (such as the capacitance from the word line to the bit line). If C_{ox} is 400 aF, $C_{parasitic} = 100$ aF, $R_{gate} = 4 \Omega$, and there are 512 bit lines in the array then the delay time through the word line is estimated as 500 ps. To fully turn the word line on (not just to the 50% point where delay is measured), we would probably want 3 ns.

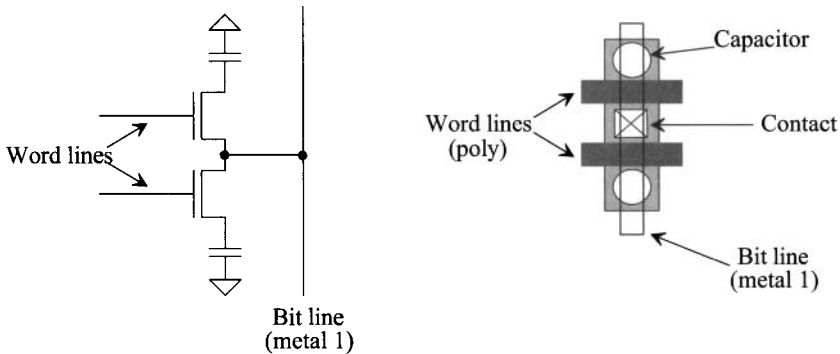


Figure 16.18 Two mbits sharing a contact to the bit line.

A section of the area layout for an open architecture DRAM is seen in Fig. 16.19. Notice that at the intersection of every bit and word line is a memory cell. A common term used to describe the density of, or distance in, a periodic array is *pitch*. As seen in Fig. 16.19, the pitch is defined as the distance between like points in the array. We used the distance between the right side of the digit lines to show pitch in this figure. A figure of merit for memory cell layout is its area. The area of the mbit DRAM cell seen in this figure is $6F^2$ where $F = \text{pitch}/2$. Question: How many MOSFETs are laid out in Fig. 16.19? Answer: Because polysilicon over active forms a MOSFET, we simply count the number of times poly crosses active (12 MOSFETs).

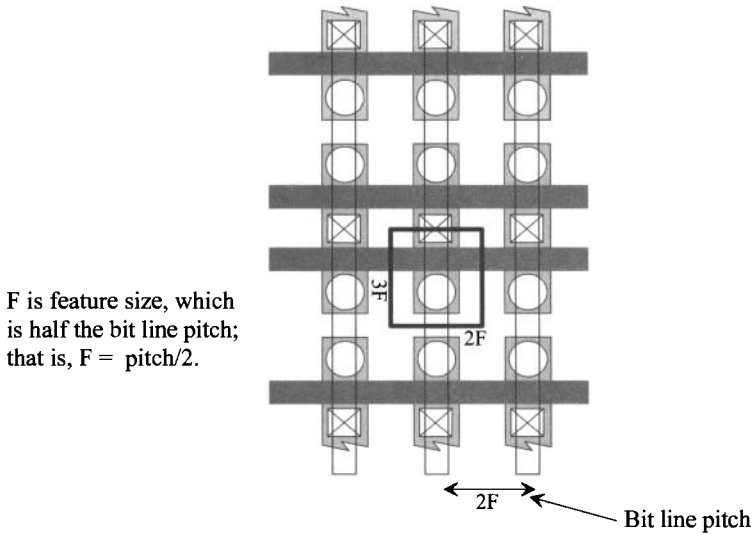


Figure 16.19 Layout of mbit used in an open bit line configuration.

Figure 16.20 shows the layout of the cell used in the folded area architecture. The schematic seen in this figure is different from the one seen in Fig. 16.17. In Fig. 16.20 our mbits share the contact to the bit line. This means that memory cells are located at the intersection of every other mbit pair with the bit lines. Figure 16.21 shows a section of the array layout in the folded architecture. The cell size is $8F^2$. The increase in the cell size is due to the needed poly interconnect between adjacent cells (the poly that runs over the field oxide, FOX).

Figures 16.22 and 16.23 show the process cross sectional views for mbits using trench capacitors and buried capacitors, respectively. The trench capacitor is formed by etching a hole in the substrate. For high-density memory, the aspect ratio (the ratio of the

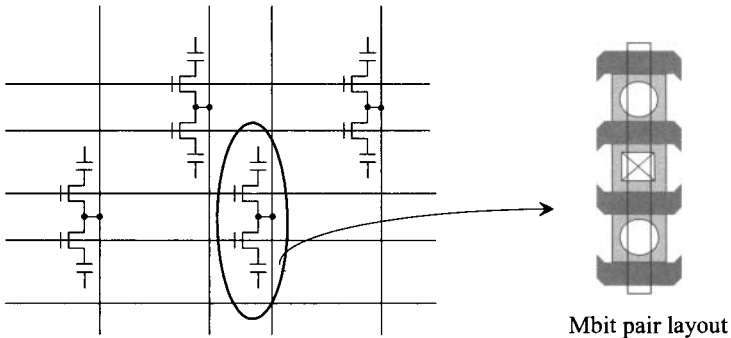


Figure 16.20 The mbit pair used for a folded architecture.

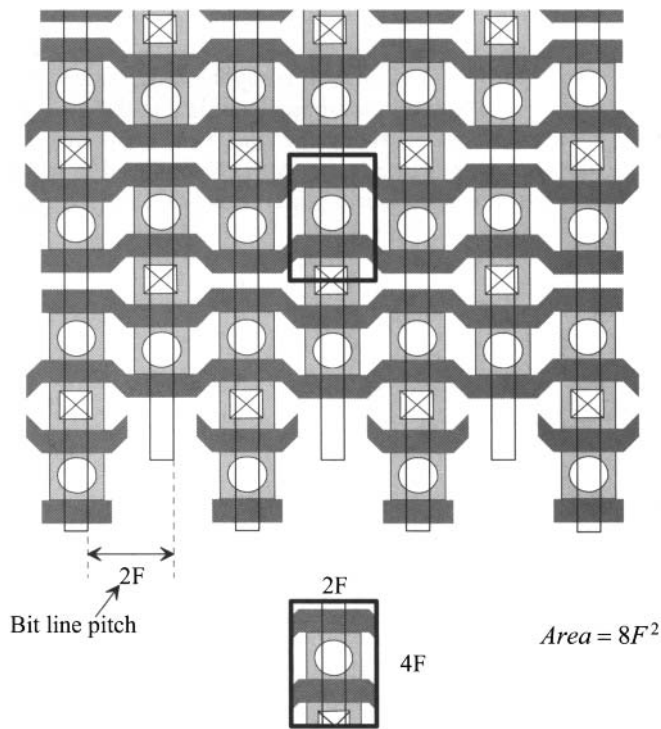


Figure 16.21 Folded architecture layout and cell size.

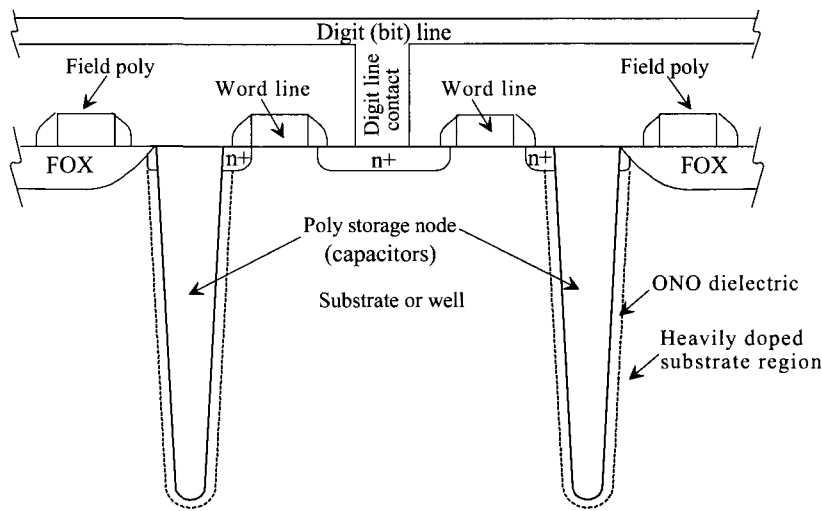


Figure 16.22 Cross-sectional view of a trench capacitor cell.